

Skills Summary

Intersections as Solutions to Systems of Two Linear Equations

Use this reference while completing your worksheet or when studying.

Skill 1 — Test Whether a Point Solves a System

Definition: (x, y) is a **solution of the system** when it makes *every* equation true.

- One equation true, the other false $\Rightarrow$ the point is on one line only — not a solution.
- Both true $\Rightarrow$ the point is on both graphs, so it is where they cross.

Substitute into each equation **separately**; never mix the two.

Example: Is $(2, 5)$ the solution of $x + y = 7$ and $y - x = 3$?

Step 1. A solution has to satisfy both equations, so test the same pair twice rather than solving anything.

Step 2. $x + y = 2 + 5 = 7 \checkmark$ and $y - x = 5 - 2 = 3 \checkmark$. Both hold, so the lines cross at $(2, 5)$

Try it: Is $(1, 4)$ the solution of $y = 3x + 1$ and $x + y = 5$?

check: $(1, 4)$ satisfies both

Skill 2 — Read an Intersection from a Graph or Table

On a graph: find the one point the two lines share and read its coordinates.

In a table: run down the two output columns and find the row where the y -values are **equal**. That x and shared y are the intersection.

If the outputs are approaching each other, the crossing is nearby; if the gap is constant, the lines are parallel and never meet.

Example: Where do $y = x + 1$ and $y = -x + 5$ meet?

Step 1. The crossing is the input that makes both rules give the same output, so scan the table for equal outputs.

Step 2. At $x = 2$ both columns read 3; at $x = 1$ they read 2 and 4, and at $x = 3$ they read 4 and 2 — the columns swap order, so they meet at $(2, 3)$

Try it: Tables for $y = x - 1$ and $y = -2x + 8$ agree at one row. Which point?

check: $(3, 2)$

Skill 3 — Solve to Find the Intersection

Why algebra works: at the crossing point both equations describe the same y for the same x , so you may set the two expressions for y equal.

- Both solved for y : set them equal, solve for x , then back-substitute for y .
- Not solved for y : isolate one variable first, then substitute.
- Always write the answer as an ordered pair and check it in *both* originals.
- A false statement such as $1 = -4$ means parallel lines: **no solution**.

Example: Solve $y = 4x - 3$ and $y = x + 3$.

Step 1. Both are solved for y , and at the intersection the two y -values are the same number, so the expressions must be equal there.

Step 2. $4x - 3 = x + 3 \Rightarrow 3x = 6 \Rightarrow x = 2$, then $y = 2 + 3 = 5$ (and $4(2) - 3 = 5 \checkmark$), so **(2, 5)**

Try it: Solve $y = 2x - 4$ and $y = -x + 5$.

(2, 3) :check:

(!) Watch out: A point that satisfies only one equation is *not* a solution — it just lies on one of the two lines. And after solving for x , do not stop: the solution of a system is a **point**, so report both coordinates.
